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Div:A

1. Write a C++ Program to Check Whether a Number is Prime or not.

```cpp
#include <bits/stdc++.h>

using namespace std;

bool isPrime(int n) {

  if (n <= 1) return false;

  for (int i = 2; i <= sqrt(n); i++) {

    if (n % i == 0) return false;

  }

  return true;

}

int main() {

  int number;

  cout << "Enter a number: ";

  cin >> number;


  if (isPrime(number)) {

    cout << number << " is a prime number." << endl;

  } else {

    cout << number << " is not a prime number." << endl;

  }


  return 0;

}
```

```
Enter a number: 50
50 is not a prime number.
```

2. Write a program to check whether the entered year is a leap year or not (a year is leap if it is

divisible by 4 and divisible by 100 or 400.)

```
#include <bits/stdc++.h>

using namespace std;


bool isLeapYear(int year) {

   if (year % 4 == 0) {

      if (year % 100 == 0) {

         if (year % 400 == 0) return true;

         return false;

      }

      return true;

   }

   return false;

}


int main(){

   int year;

   cout<<"enter a year:";

   cin>>year;


if (isLeapYear(year)){

   cout<<year<<" is a leap year."<<endl;

}

else{

   cout<<year<<" is not a leap year."<<endl;

}
```

```
}
```

```
enter a year:2026
2026 is not a leap year.
```

3. Write a program to find the factorial of a number.

```
#include <bits/stdc++.h>
using namespace std;

int factorial(double n) {
   if (n == 0) {
      return 1;
   } else {
      return n * factorial(n - 1);
   }
}

int main() {
   double number;
   cout << "Enter a number: ";
   cin >> number;

   if (number < 0) {
      cout << "Factorial is not defined for negative numbers." << endl;
   } else {
      cout << "Factorial of " << number << " is " << factorial(number) << "." << endl;
   }

   return 0;
```

```
}
```

```
Enter a number: 10
Factorial of 10 is 3628800.
```

4. Write a program to check if the number is Armstrong or not.

```
#include <bits/stdc++.h>
using namespace std;

int digitCount(int n) {
  int count = 0;
  while (n > 0) {
    n /= 10;
    count++;
  }
  return count;
}

int power(int base, int exp) {
  int result = 1;
  for (int i = 0; i < exp; i++) {
    result *= base;
  }
  return result;
}

bool isArmstrong(int n) {
  int original = n;
  int sum = 0;
  int digits = digitCount(n);
```

```
    while (n > 0) {
        int digit = n % 10;
        sum += power(digit, digits);
        n /= 10;
    }

    return sum == original;
}

int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;

    if (isArmstrong(number)) {
        cout << number << " is an Armstrong number." << endl;
    } else {
        cout << number << " is not an Armstrong number." << endl;
    }

    return 0;
}
```

```
Enter a number: 45
45 is not an Armstrong number.
```

5. Write a C++ program to find the Fibonacci series till the limit entered by the user

```
#include<bits/stdc++.h>
```

```
using namespace std;

void fibonacci(int n){
  int a=0,b=1,c;
  for(int i=0;i<n;i++){
    cout<<a<<" ";
    c=a+b;
    a=b;
    b=c;
  }
}

int main(){
  int n;
  cout<<"Enter the limit for Fibonacci series: ";
  cin>>n;

  cout<<"Fibonacci series till "<<n<<" terms: ";
  fibonacci(n);
  cout<<endl;

  return 0;
}
```

```
Enter the limit for Fibonacci series: 6
Fibonacci series till 6 terms: 0 1 1 2 3 5
```

. 6. Write a C++ program that finds the largest among three different numbers entered by the user.

```
#include <bits/stdc++.h>
using namespace std;

int findLargest(int a, int b, int c) {
  return max(a, max(b, c));
}

int main() {
  int num1, num2, num3;
  cout << "Enter three numbers: ";
  cin >> num1 >> num2 >> num3;

  int largest = findLargest(num1, num2, num3);
  cout << "The largest number is: " << largest << endl;

  return 0;
}
```

```
Enter three numbers: 1 5 89
The largest number is: 89
```

7. Write a program in C++ to display n terms of natural numbers and their sum. Sample Output:

Input a number of terms: 7

The natural numbers upto 7th terms are: 1 2 3 4 5 6 7

The sum of the natural numbers is: 28

```
Enter the number of terms: 10
The natural numbers up to 10 terms are: 1 2 3 4 5 6 7 8 9 10
The sum of the natural numbers is: 55
```

8. Write a C++ program that performs basic arithmetic operations (addition, subtraction,

multiplication, division) using user-defined functions.

```cpp
#include<bits/stdc++.h>

using namespace std;


void Operations(int a, int b) {

    cout << "Addition: " << a + b << endl;

    cout << "Subtraction: " << a - b << endl;

    cout << "Multiplication: " << a * b << endl;

    if (b != 0) {

        cout << "Division: " << static_cast<double>(a) / b << endl;

    } else {

        cout << "Division: Cannot divide by zero." << endl;

    }

}


int main() {

    int num1, num2;

    cout << "Enter two numbers: ";

    cin >> num1 >> num2;


    Operations(num1, num2);


    return 0;

}
```

}

```
Enter two numbers: 45 65
Addition: 110
Subtraction: -20
Multiplication: 2925
Division: 0.692308
```

9. Write a C++ program to calculate the factorial of a given number using a user-defined function.

```cpp
#include <bits/stdc++.h>

using namespace std;


int factorial(int n) {

  if (n == 0) {

    return 1;

  } else {

    return n * factorial(n - 1);

  }

}


int main() {

  int number;

  cout << "Enter a number: ";

  cin >> number;


  int result = factorial(number);

  cout << "Factorial of " << number << " is " << result << "." << endl;


  return 0;

}
```

```
Enter a number: 15
Factorial of 15 is 2004310016.
```